Robert Ondino Morelli

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Education

B.S. Computer Science *[University of Utah]* *May 2027*

Technical skills

Zig, C/C++, C#, Python, Java, TS, WASM, Git, Docker, GitHub Actions, CI/CD

Experience

Research Assistant *[University of Utah]* *Oct 2025 – Present*

- Analyzing performance cost of gradual typing in Meta's Static Python (CinderX)
- Automatic rewriting of Python code to construct test examples
- First author, "Automatic Cast Insertion to Evaluate Progressive Gradual Types" (under review)

Teaching Assistant *[University of Utah]* *Sep 2025 – May 2026*

- Led labs, graded, and assisted students for COMP 1020

Software Engineer/DevOps *[Stutor Inc.]* *Sep 2023 – Apr 2024*

- Architected automation pipeline using Fastlane and GitHub Actions to build, sign and deploy to multiple platforms
- Added compound keys to tables where backend queries had drifted from database schema; query times reduced from 15+ s to <1 s

Web Developer/DevOps *[Jerran Software Solutions, consulting]* *Apr 2022 – Sep 2023*

- Architected, proposed and implemented overhaul of development process for the LDS MTC team using build automation in conjunction with new QA practices; deployment was reduced from a week-long headache to mere minutes
- Rewrote Embark startup using dependency-ordered queue for async initializations; first-time loading reduced by 50%

Research Assistant Intern *[Earl Keefe PhD, University of Utah]* *Nov 2020 – Jun 2021*

- Used pypng to construct novel visualizations of covariance matrices to provide more intuitive understanding of data

Projects

IRX regex fuzzer *[Zig]* ↗

- Fuzzes patterns instead of inputs: samples regexes that must match one fixed string and must reject another fixed string

UTU language and compiler *[JS/TS, WasmGC]* ↗

- WasmGC language with predictable lowering, small bundles, and .d.ts as its ABI

Spreadsheet formulas to DLL compiler *[C#]* ↗

- Compiles spreadsheet formulas into a DLL, supporting optimizations including fast-math style unsafe float conversion

Open source & competitions

Contributed to open-source IDE code-base *[MARS]* ↗

- Discovered and corrected oversight in the MARS IDE's instruction decoder that affected correctness of self-modifying code

CinderX bug hunting *[CinderX]*

- Reported 11 bugs in CinderX and contributed 2 fixes to the type-checker

Regex bug hunting

- Reported 86 correctness bugs across a wide variety of regular expression engines including Perl, .NET and V8

Competitive programming *[ICPC]* ↗

- As part of team "Big Cottonwood Coders", placed 12 out of 48 teams at the Rocky Mountain regional contest (2025)
- As Event Coordinator of the U of U technical coding club, organized and oversaw competitions (2026)